

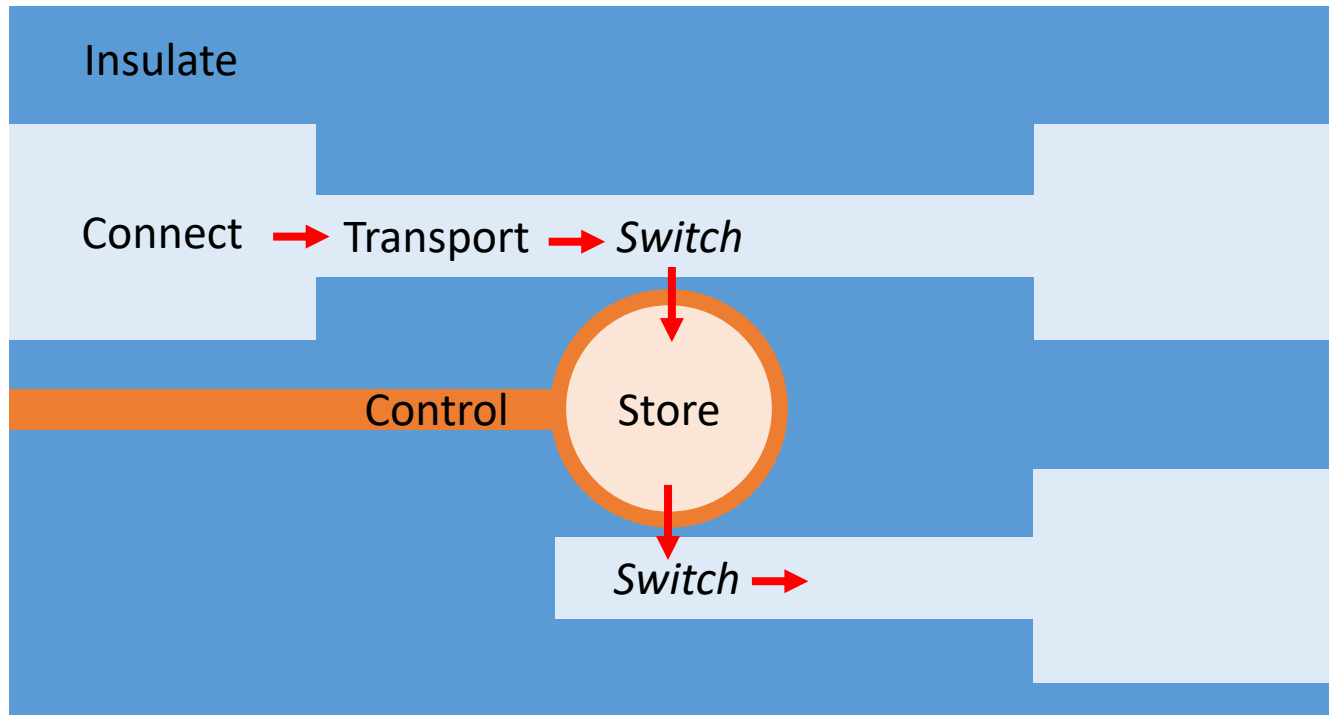
Plasmonic materials and devices

- **Circuit model**
- **Drude model (revision)**
- Localized Surface Plasmons
- Surface Plasmon Polaritons



Optical circuit analogy

Computers use miniature circuits of insulators, semiconductors & conductors



The size of photonic circuits (μm) is large compared to electronic ($0.01\mu\text{m}$)

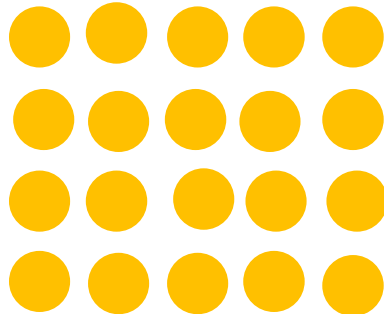
A major research goal is to combine **photons + electrons = plasmonics**

There are some obvious differences that have to be overcome

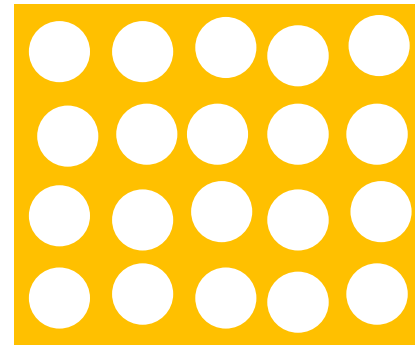
Insulator crystals

Just use electrical insulators

There *are* structures analogous to crystals – called “plasmonic **metamaterials**”
However their function is not really as insulators



Mostly transmitting at low frequencies



Conductor (diluted)

Mostly reflecting at low frequency

Both can have (very) interesting resonances
Primarily determined by individual resonators

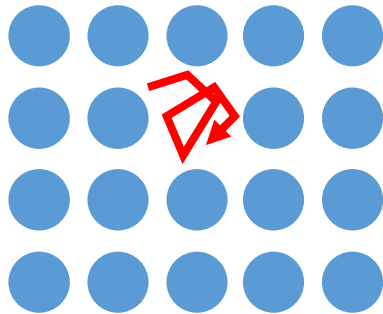
The behaviour of plasmonic metamaterials is complicated
We may return to this at a later stage

Resonator

To trap information

Different/absent inclusion

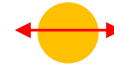
Photonic resonator



low index

Particle or disc

Plasmonic resonator



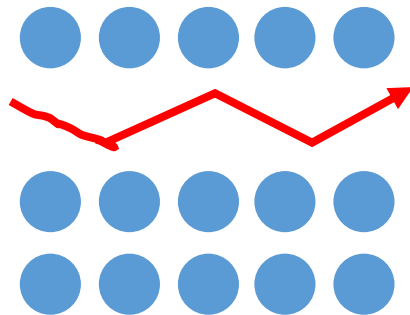
Localized Surface Plasmon (LSP)

Conducting defect

To transport information

Line of missing inclusions

Photonic waveguide



Plasmonic waveguide

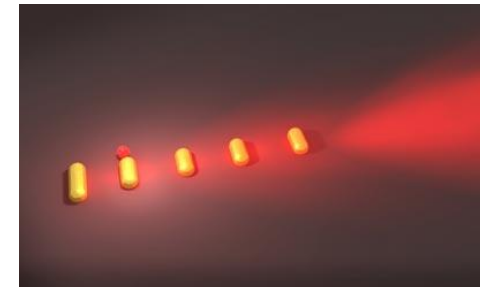
film/strip-line



Surface Plasmon Polariton (SPP)



Optical circuits



Antenna – wireless

Conductors have highly dispersive response

Let's review some of the key points of plasma dispersion

Drude model

http://en.wikipedia.org/wiki/Drude_model

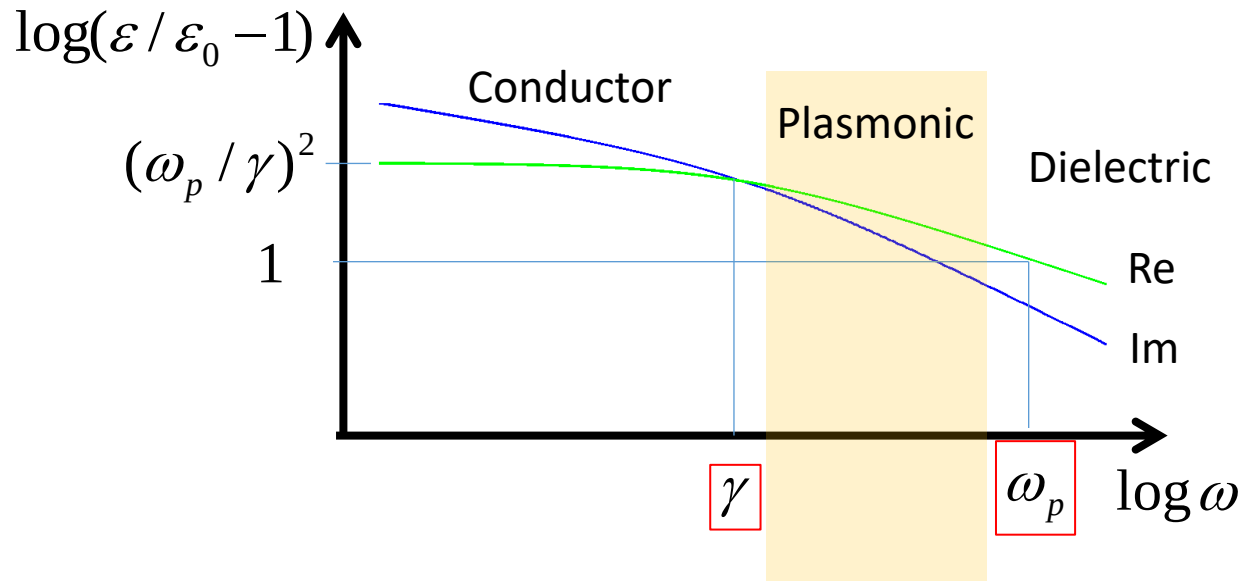
Plasma = nearly free electrons damped by interactions

Exists in conductors (metals AND doped semiconductors)

$$\epsilon / \epsilon_0 - 1 = \frac{-\omega_p^2}{\omega^2 + i\omega\gamma}$$

← **Plasma frequency $\propto$ charge density** $\omega_p^2 = Ne^2 / (m\epsilon_0)$

← Damping frequency $\gamma = \epsilon_0 \omega_p^2 \rho_{DC}$



Negative permittivity
Some imaginary = lossy

Let's do some examples of plasma parameters for conductors

noting that Drude is generally a gross simplification, especially near plasma frequency.

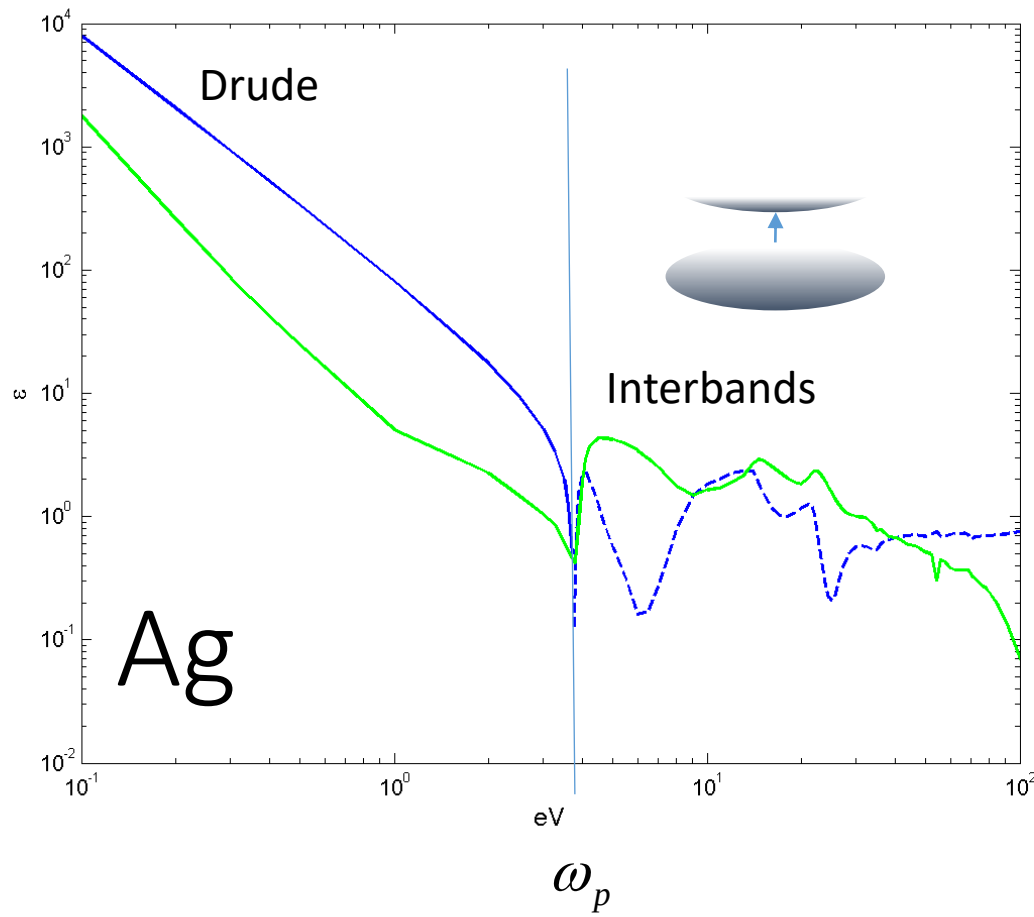
Material	Carrier conc cm^{-3}	DC resist $\Omega \text{ cm}$	Plasma μm	Damping μm
<i>Al</i>	1.8×10^{23}	3×10^{-6}	0.08	10^1
<i>K</i>	1.4×10^{22}	7×10^{-6}	0.3	7×10^1
low carrier-density conductors (e.g. TiN, ITO, VO_2) their properties vary depending on preparation				
<i>Si heavy</i>	10^{20}	10^{-3}	3	7×10^1
<i>Si undoped</i>	10^{12}	10^4	3×10^4	7×10^2

$$\lambda_p = 2\pi c / \sqrt{Ne^2 / (m\epsilon_0)}$$

$$\lambda_\gamma = \lambda_p^2 / (\epsilon_0 \rho_{DC} 2\pi c)$$

Drude vs real material

In many real materials the “high” frequency approximation is only good when $\omega \ll \omega_p$



- *Estimated* plasma frequency for Ag is 9eV (140nm)
- *Observed* frequency is 4eV (340nm): due to interbands

Drude model limits

Let's look at some approximations to simplify later derivations

$$\varepsilon / \varepsilon_0 - 1 = \frac{-\omega_p^2}{\omega^2 + i\omega\gamma}$$

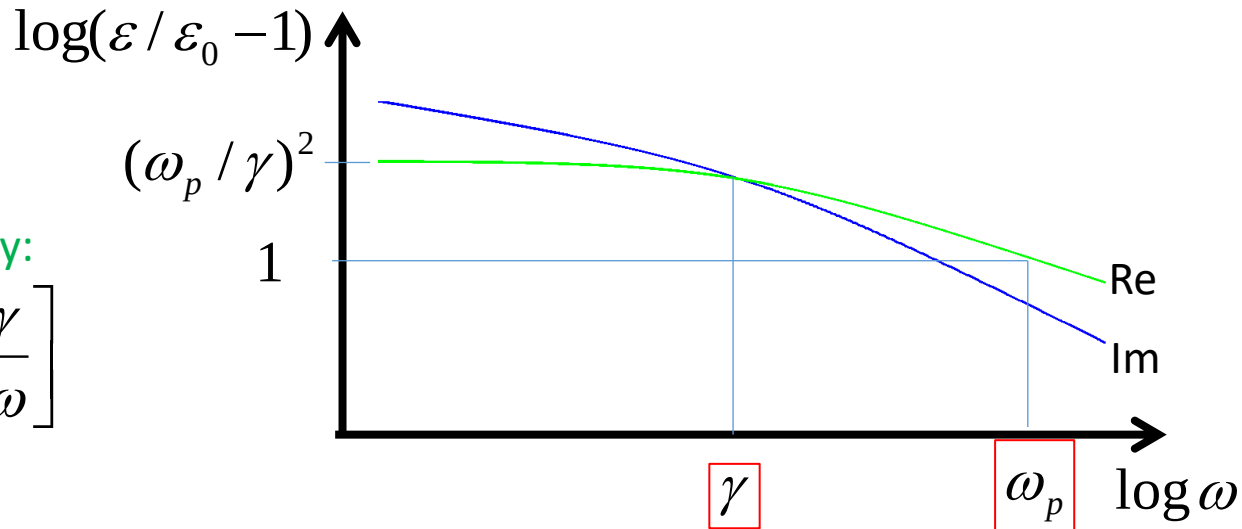
Splitting into real and imaginary:

$$\varepsilon / \varepsilon_0 - 1 = \frac{\omega_p^2}{\omega^2 + \gamma^2} \left[-1 + i \frac{\gamma}{\omega} \right]$$

Now the approximations...

$$\text{Low } \omega \ll \gamma: \quad \varepsilon / \varepsilon_0 - 1 \sim \frac{\omega_p^2}{\gamma^2} \left[-1 + i \frac{\gamma}{\omega} \right]$$

$$\text{High } \omega \gg \gamma: \quad \varepsilon / \varepsilon_0 - 1 \sim \frac{\omega_p^2}{\omega^2} \left[-1 + i \frac{\gamma}{\omega} \right]$$



Imag part dominates (currents)

Real part dominates (displacement field)

In many real materials the “high” frequency approximation is only good when $\omega \ll \omega_p$

Nevertheless Drude is generally ok at moderate frequency and is easy to write down

Plasmons

Plasma = gas of free charge carriers, e.g. the carriers in a conductor

‘on’ = wavicle

A surface plasmon is a

- collective oscillation of charge carriers near a surface
- a quasiparticle hybrid between electron and photon
- an evanescent wave on a conducting surface...

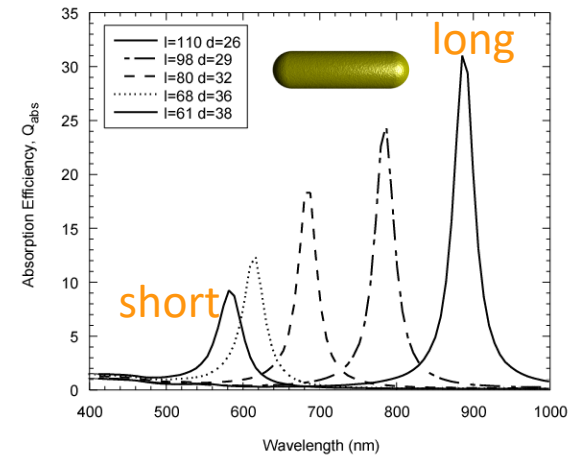
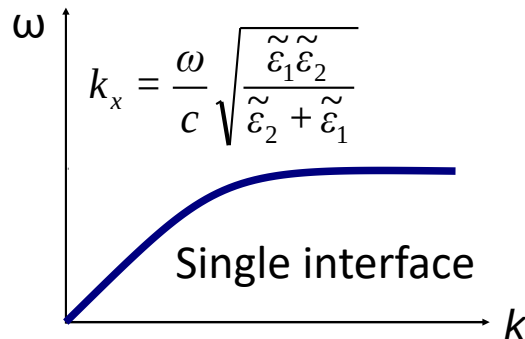
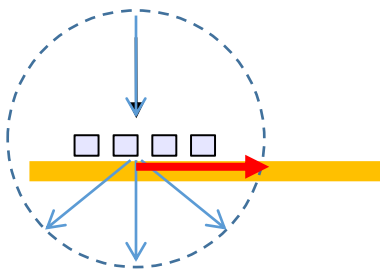
There are two limiting cases of plasmons

- ❖ **LSP** = localized surface plasmon (e.g. on a small particle)
- ❖ **SPP** = surface plasmon polariton (e.g. on a metal film)

The distinction can become less obvious in-between these two cases

Summary

- Plasmons = surface resonance of conductor
- Drude model useful for basic material response
- LSP: small particle traps
 - Shape most useful for tuning
 - Electron forces key to understanding
 - Modelled by quasistatic resonance
- SPP: thin film waveguides
 - Needs special coupling (prism/gratings)
 - Highly dispersive: light-like → slow light
 - Thickness determines properties



$$\alpha = V \sum_m \frac{A_m}{S_m + 1/\chi}$$

